Answer key. Answers are in magenta.

NRES 746, Week 8, Lecture 1: How Much Did the Prior Matter?

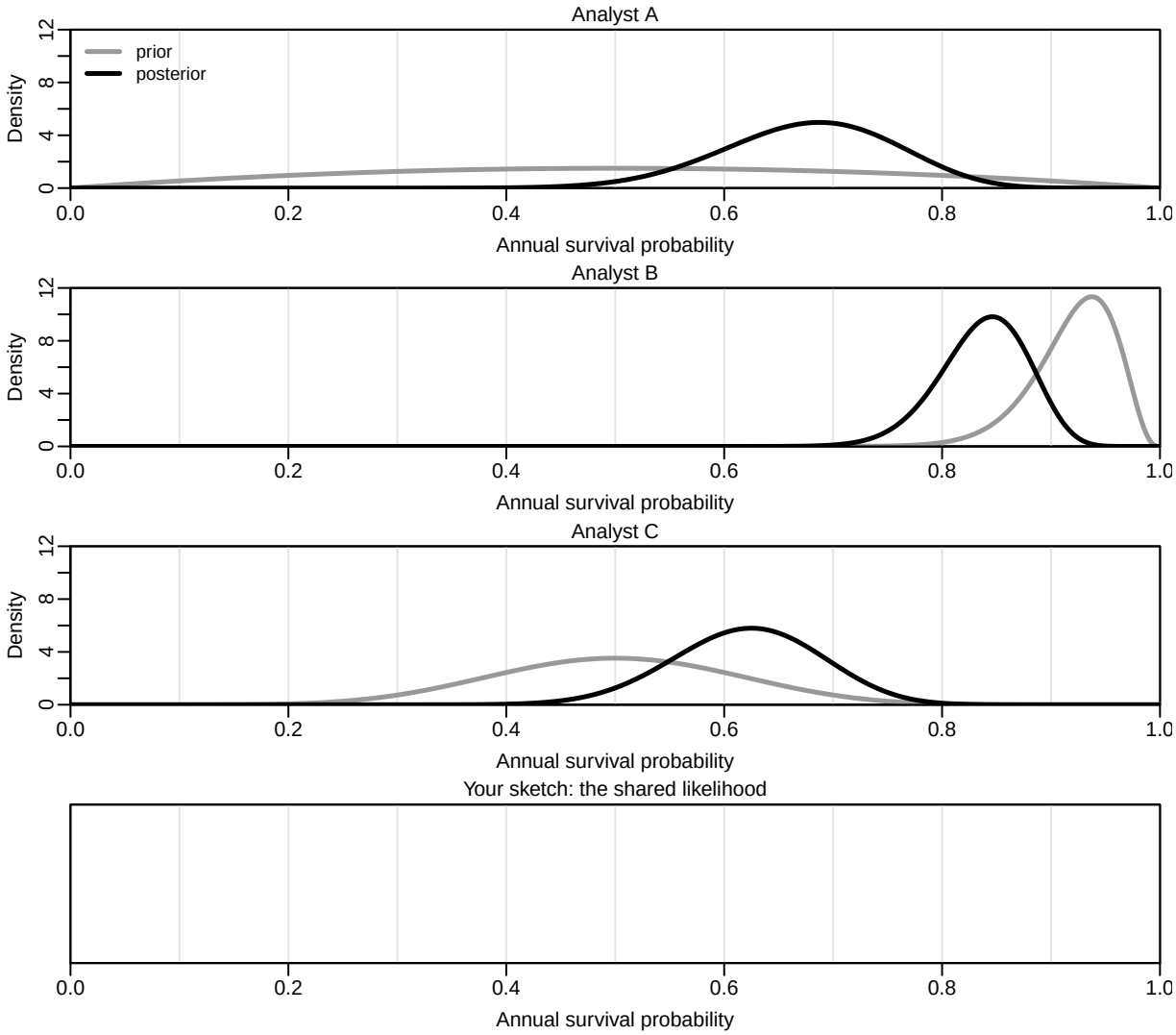
Instructions: Work alone on Part 1 (about 4 minutes), then compare with a neighbor for Part 2. Keep laptops closed.

The study

A biologist radio-tracks juvenile desert tortoises at one site and records which ones survive the year. Three analysts estimate annual survival from the **same data**, but each chooses a different prior:

- **Analyst A** uses a weak prior centered on 0.5.
- **Analyst B** uses earlier estimates from juveniles raised in a predator-proof pen.
- **Analyst C** uses an estimate from a severe drought year at another site.

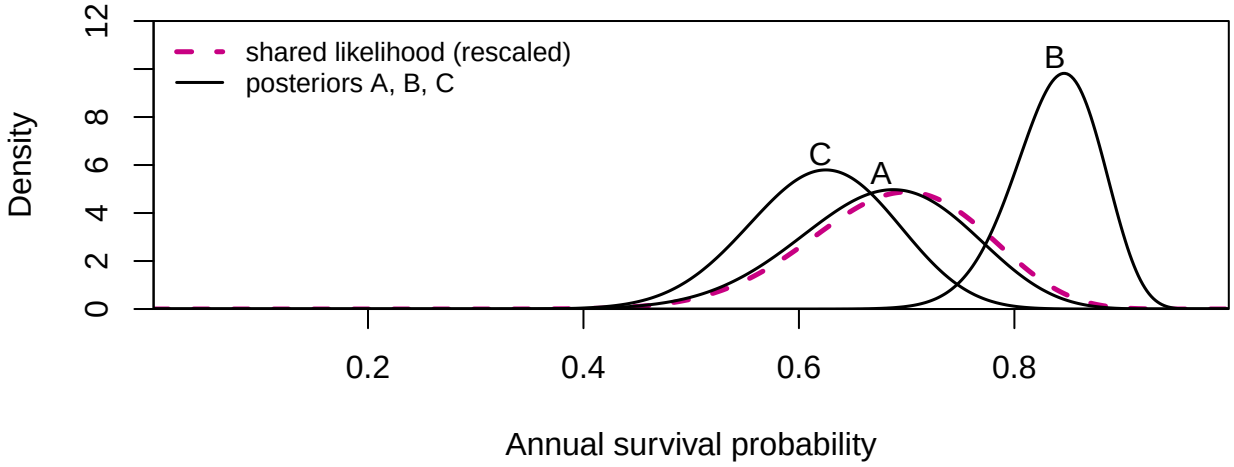
Each panel shows one analyst's prior (grey) and posterior (black). The likelihood, which is the same in all three panels, is not shown.



Part 1: Find the likelihood (individual)

1a. Sketch the shared likelihood on the bottom panel (the height doesn't matter; get its peak and width roughly right). Which analyst's panel did you rely on most, and why?

A curve peaking near 0.70, about as wide as Analyst A's posterior (dashed magenta below). Rely on Analyst A: with a nearly flat prior, A's posterior is almost the likelihood itself.



1b. Rank the three analysts' results from most to least dependent on their choice of prior. Give a one-line reason for each.

B (most): pulled far toward 0.92; its prior is worth 50 tortoises vs. 30 in the data. C: pulled partway toward 0.50 (prior worth 20). A (least): nearly matches the likelihood (prior worth 4).

Part 2: Which number? (with a neighbor)

2a. The three posterior means for annual survival are 0.68 (A), 0.84 (B), and 0.62 (C). A manager needs one survival rate for a population model. What would you tell them, and what would you report?

Sample answer: don't use B, since a predator-proof pen says little about survival in the wild. Report A (mean 0.68, 95% credible interval 0.51 to 0.82), and note that a drought-year prior (C) pulls it down to 0.62; 30 tortoises aren't enough to make the prior irrelevant.